

Los números de las simulaciones de Bernardo vienen dados por el Stokes number

$$St = \tau_p / \tau_\eta$$

Maxey-Riley entero ([Angriman et al. \(2020\)](#)):

based on the equations for the dynamics of inertial neutrally-buoyant point particles. In principle, sufficiently small inertial particles with a density mismatch to that of the fluid can be modeled by the Maxey-Riley equation [32, 37], which when considering only first order effects in the particle radius reads

$$\frac{d\mathbf{v}}{dt} = \frac{1}{T_p} [\mathbf{u}(\mathbf{x}_p, t) - \mathbf{v}(t)] + \frac{3}{2} \frac{\gamma}{1 + \gamma/2} \frac{D\mathbf{u}(\mathbf{x}_p, t)}{Dt} + \frac{1 - \gamma}{1 + \gamma/2} \mathbf{g}, \quad (9)$$

where $\mathbf{v}(t)$ is the particle's velocity, $\mathbf{u}(\mathbf{x}_p, t)$ is again the fluid velocity at the particle position, T_p is the Stokes time defined as in Eq. (2), $\gamma = \rho/\rho_p$ is the fluid to particle density ratio, D/Dt is the convective derivative, and $\mathbf{g}$ is the acceleration of gravity. This approximation is valid for

Maxey-Riley sin término de historia ([Español et al. \(2025\)](#)):

neutrally buoyant particles we solve the Maxey-Riley-Gatignol equation (Maxey & Riley 1983; Gatignol 1983), neglecting the Basset-Boussinesq term (we explicitly verified that preserving this term yields similar results in the analysis below),

$$\frac{d\mathbf{v}}{dt} = \frac{1}{\tau_p} [\mathbf{u}(\mathbf{x}, t) - \mathbf{v}(t)] + \frac{D}{Dt} \mathbf{u}(\mathbf{x}, t), \quad (2.6)$$

where $\tau_p = a^2/(3\nu)$ is the particle response time, a is the particle radius, $\mathbf{v}(t)$ is the particle velocity, and $\mathbf{u}(\mathbf{x}, t)$ is the fluid velocity at the particle position. For big

Only non-linear drag ([Español et al. \(2025\)](#)):

En el segundo set de simulaciones de Bernardo NLD=ONLD.

En el primero NLD tenía el término de aceleración extra.

En GHOST hay un γ extra en Re_p que da el mass ratio, pero siempre está en 1. Lo único que cambiamos de las partículas es el radio.

to their dynamics (Reijtenbagh *et al.* 2023). We therefore consider instead a simplified model of small heavy particles only with nonlinear drag (Wang & Maxey 1993; Stout *et al.* 1995) and with negligible settling velocity (note that we also neglect added mass forces (Brandt & Coletti 2022)),

$$\frac{d\mathbf{v}}{dt} = \frac{1 + 0.15Re_p^{0.687}}{\tau_p} [\mathbf{u}(\mathbf{x}, t) - \mathbf{v}(t)], \quad (2.7)$$

where $Re_p = (18\tau_p/\nu)^{1/2} |\mathbf{u}(\mathbf{x}, t) - \mathbf{v}(t)|$ is the instantaneous Reynolds number of the particle. Equation (2.6) is valid for small particles with $Re_p < 1$, while Eq. (2.6) is expected to hold, at least, for $1 < Re_p < 40$ (Wang & Maxey 1993).

Naive heavy particles ([Angriman et al. \(2020\)](#)):

DNSs. The equation for the evolution of the inertial particles in our DNSs is then given by

$$\frac{d\mathbf{x}_p}{dt} = \mathbf{v}(t), \quad \frac{d\mathbf{v}}{dt} = \frac{1}{\tau_p} [\mathbf{u}(\mathbf{x}_p, t) - \mathbf{v}(t)]. \quad (10)$$